Tracked arm60 — simulation report

Digital prototype R3 with Track R4. These results describe the supplied CAD and explicit numerical models. They do not establish a tested physical payload or certify printed strength.

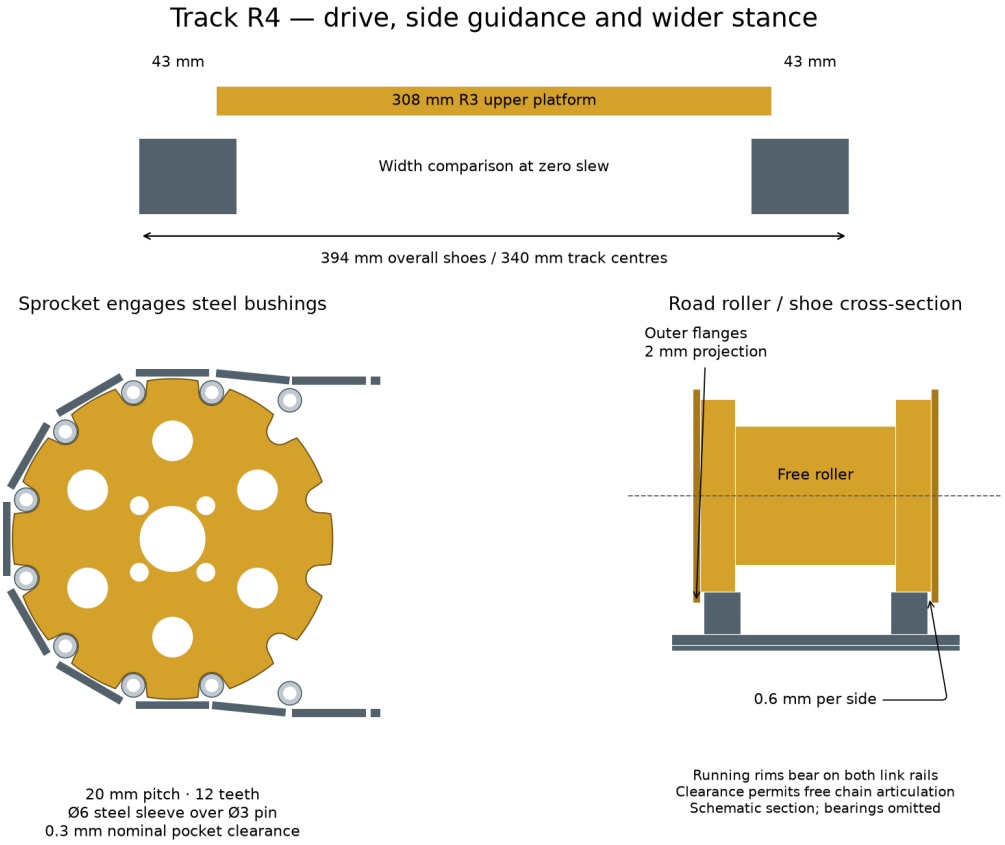


CAD and motion

Assembly intersections: 1355; cross-system intersections: 1785. Collisions above tolerance: 0. Gear cycle: 61 samples, maximum overlap 0 mm³. Track pitch closure error 1.78e-14 mm.

Sampled exact CAD intersections within the collision reference envelope and quasi-static mechanics, not FEA, continuous collision proof or physical certification. This PASS applies to that sampled subset, not the full independent motion range. Low-boom sideways poses can collide with tracks; operators manage clearance. The preview does not enforce the reference envelope.

Track R4 mechanical changes

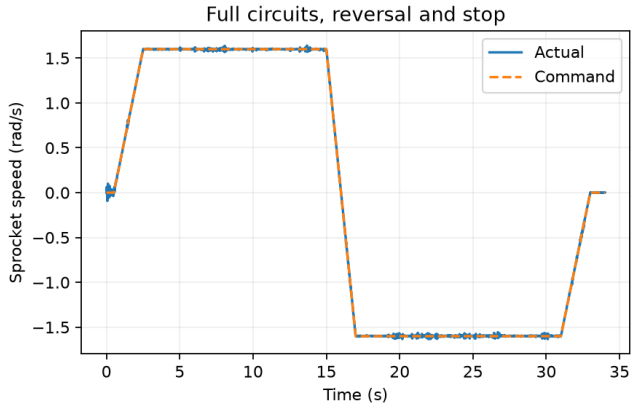


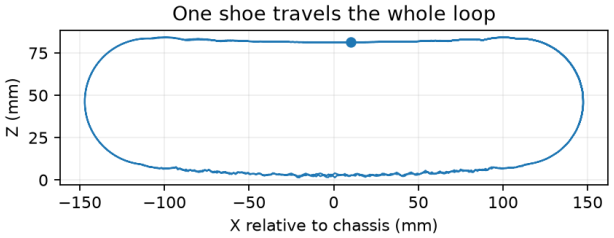
Adapted from Berco sprocket, track-chain, roller and idler references. Rounded tooth pockets drive replaceable steel bushings; outer guide flanges retain lateral running clearance. The 394 mm shoe envelope extends 43 mm beyond each side of the 308 mm R3 upper platform. See references/TRACK_SYSTEM.md for original manufacturer URLs and images.

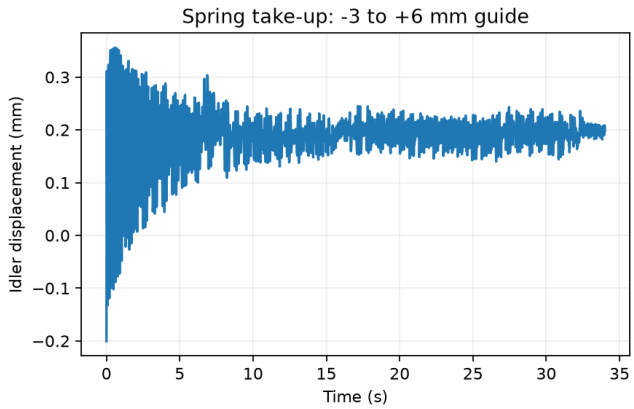
Driven track contact simulation

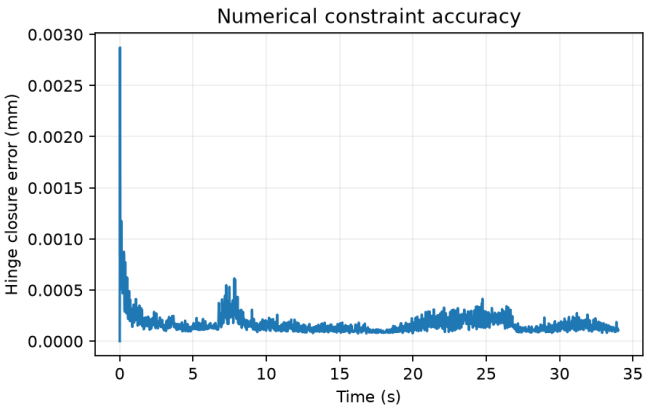
Track contact test	Result
Engine / status	MuJoCo 3.3.7 Newton / PASS
Duration / integration	34 s / 68,000 steps at 0.5 ms
Links / hinge point constraints	34 / 68
Forward / reverse mean speed	1.5982 / -1.5981 rad/s (command ±1.6)
Forward rotation / total rotation	22.371 / 47.150 rad
Speed after controlled stop	0.002103 rad/s
Peak applied torque / limit	0.201 / 1.02 N m
Shoe path / ideal pitch travel	1851.5 / 1801.0 mm; ratio 1.028
Maximum hinge / lateral error	0.002871 / 0.621917 mm
Idler displacement	-0.201 to 0.356 mm (guide -3 to +6)
Sprocket contact pieces present	114
Alternating side load / maximum lateral movement	±5 N / 0.622 mm
Left / right guide-contact observations	239 / 729
Minimum complete circuits, forward / reverse	1.14 / 1.26

Gravity-loaded fixed single-track test stand; no floor. Torque-limited motor, steel-bushing sprocket, sprung idler with rail rims, rollers with outer guide flanges, 34 free shoes and 68 hinge point constraints. Simplified rigid shoe contacts and ideal bearings; rail/frame collisions, soil, wear, printed elasticity and motor heating are not modeled.







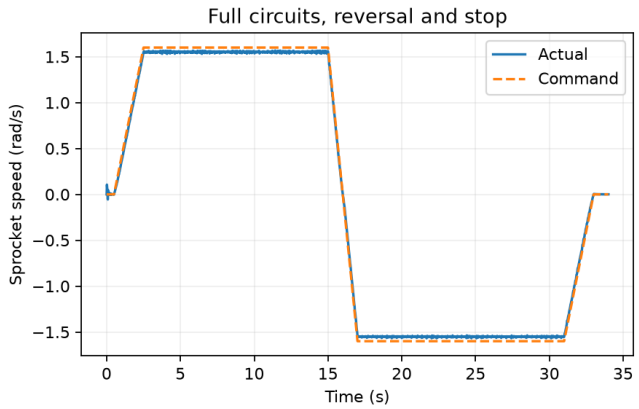


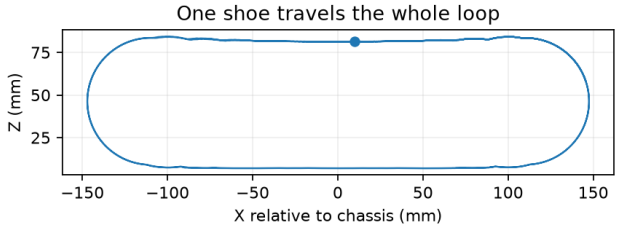
Loaded floor-contact simulation

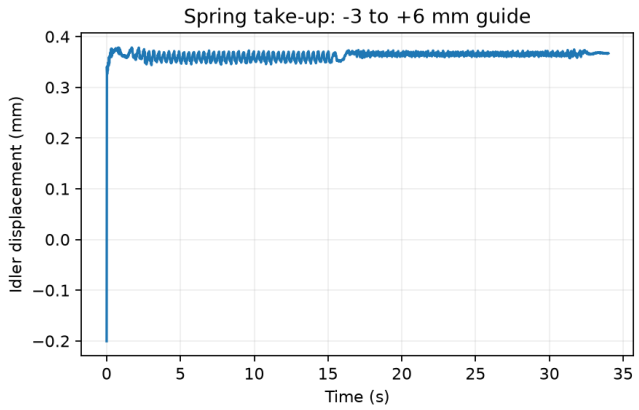
Half-machine mass on a planar carriage free in X/Z, rigid floor and actual shoe contact; pitch/roll/yaw guided. Torque-limited motor, steel-bushing sprocket, sprung idler with rail rims, rollers with outer guide flanges, 34 free shoes and 68 hinge point constraints. Simplified rigid shoe contacts and ideal bearings; rail/frame collisions, soil, wear, printed elasticity and motor heating are not modeled.

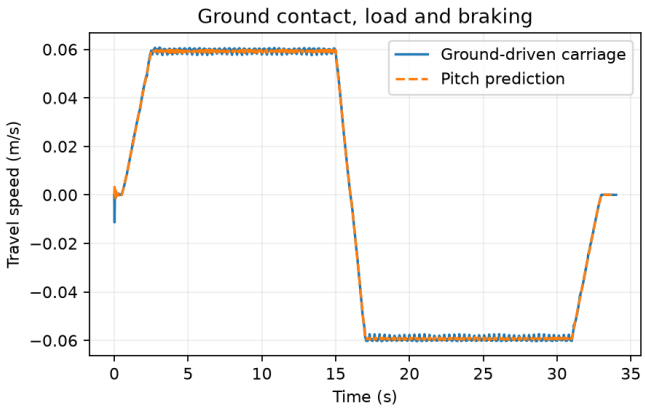
Actual CAD checked at 36 recorded track poses: 2446 exact intersections, no overlap above 0.5 mm3. This allowance covers soft contact and polygonal mesh approximation; these are sampled checks.

Loaded track test	Result
Status	PASS
Supported mass / weight	6.022 kg / 59.08 N
Mean floor reaction	59.05 N
Road roller share of supported weight	98.7%
Road roller upward reactions (rear / centre / front)	19.38, 19.70, 19.21 N
Carrier downward reactions (rear / front)	0.188, 0.192 N
Passive wheel forward angular speeds	1.56, 3.29, 3.29, 3.29, 4.21, 4.21 rad/s
Passive wheel reverse angular speeds	-1.56, -3.28, -3.28, -3.28, -4.22, -4.20 rad/s
Forward / reverse travel	0.0594 / -0.0594 m/s
Pitch prediction at commanded speed	0.0611 m/s
Final travel speed	0.000028 m/s
Minimum complete circuits, forward / reverse	1.12 / 1.20
Maximum hinge error	0.001530 mm









Track references: Berco drive sprockets, rollers, track chains and idlers; Caterpillar 320 technical specifications, pages 2, 3, 4 and 20. See references/TRACK_SYSTEM.md for source dimensions, model adaptations and simulation scope. The offline

viewer interpolates the recorded rigid-body poses; it does not run a live terrain solver.

Stability and load requirements

PETG at 1.27 g/cm³ on exact solid volume; steel 7.85; PET glazing 1.38. Motor masses 250/90 g, two 18650 cells 48 g each (Panasonic NCR18650B maximum 48.5 g) plus pads and nickel strip, BMS/converters 72 g, bearings 12/70 g, U60 internals 35 g each. 1.527 kg steel ballast plate inside the counterweight. Mass, friction, gearbox thrust and printed strength must be measured. No soil digging load or dynamic arm load included in actuator thrust.

Mass 11.894 kg. Minimum calculated support margin 43.19 mm over 18954 configurations. Worst case: strokes [7.5, 0.0, 0.0], slew 150°, 0.15 kg load.

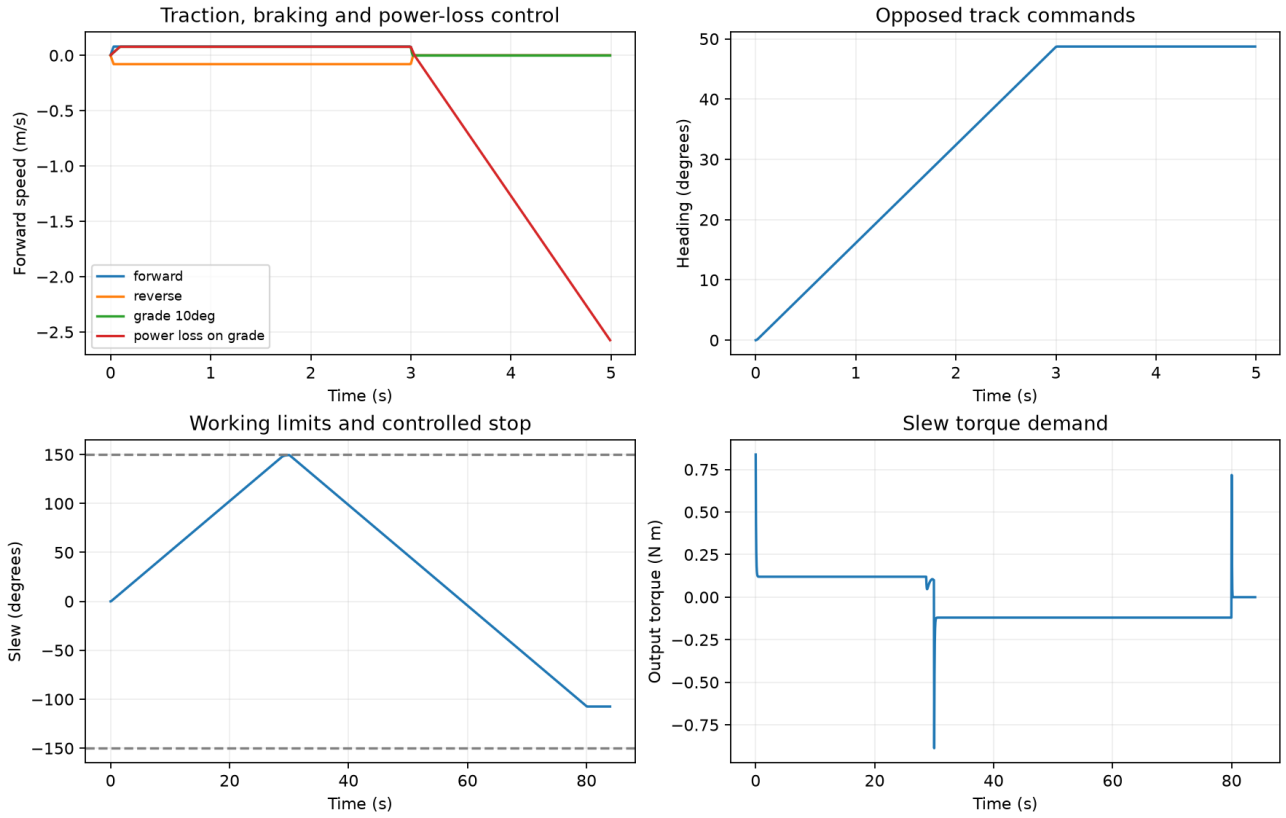
Slew overturning moment 2.93 N m; bearing-couple reaction 73.3 N; steel spindle bending estimate 3.73 MPa.

Per-cylinder static thrust, boom/stick/bucket: 42.4, 45.6, 16.4 N. Select and bench-test against twice these values. These calculations omit soil reaction and dynamic arm loads.

Grade torque per drive 0.572 N m; skid-turn torque 0.389 N m. Required continuous motor rating 1.2 N m. The assumed friction coefficient is 0.30.

Simplified Lewis gear stress 3.65 MPa against an assumed printed allowable of 8 MPa; pin bending 87.6 MPa against assumed steel yield 200 MPa. These are analytic estimates, not FEA or material coupon results.

Vehicle and slew dynamics



Reduced rigid-body differential traction with force/current saturation, Coulomb scrub and rolling loss. Flat floor is the operating assumption; grade cases test the model and reveal power-off rollback. Slew assumes a raised boom clear of the tracks, ideal gears and 0.50 kg m² equivalent inertia. Motor performance assumes a maintained regulated 12 V rail within the commissioned power budget. Not full terrain contact, motor thermal analysis or firmware validation.

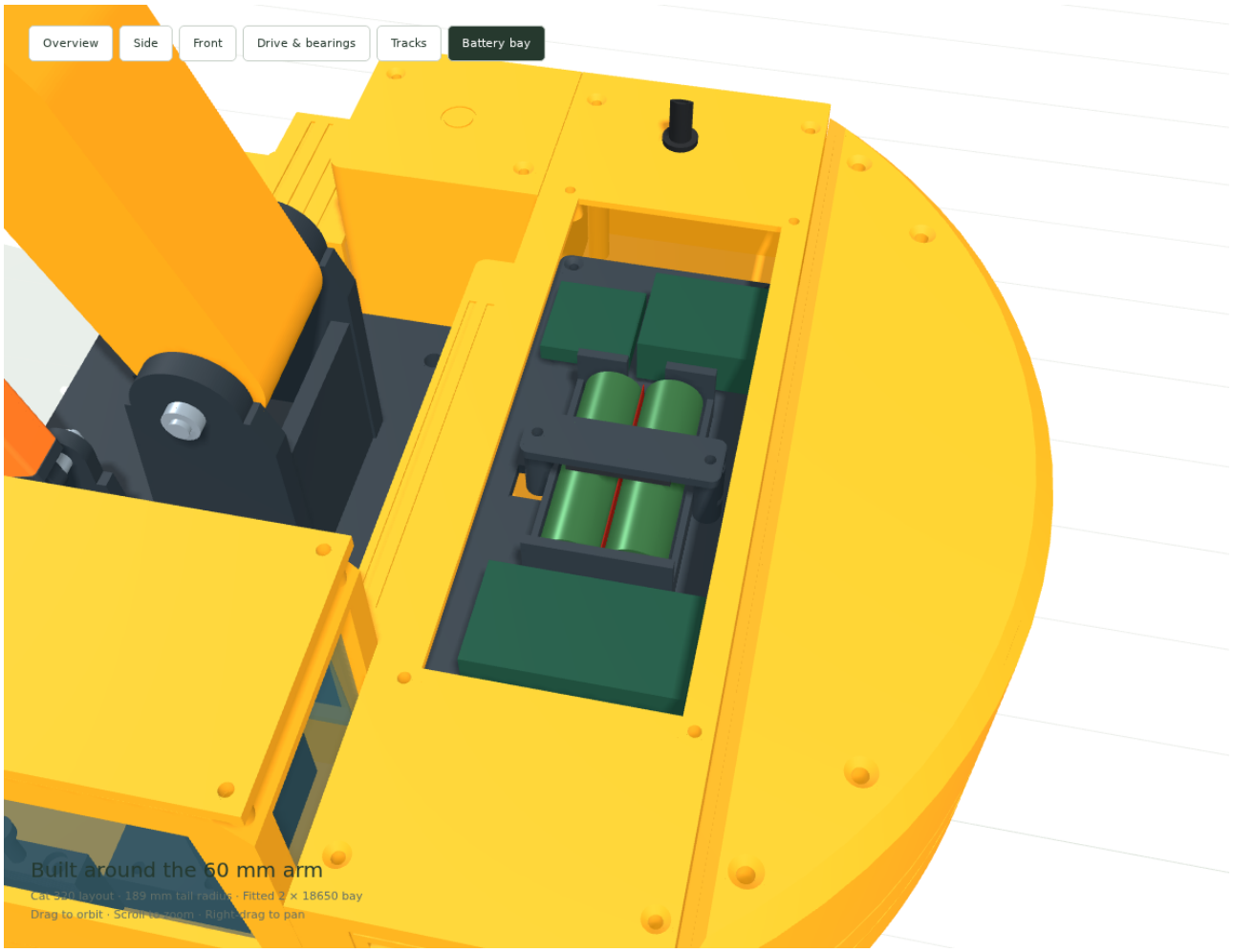
Drive shaft/bearing upper bound: outer bearing reaction 334.2 N; steel shaft bending 93.0 MPa. Specify 608 static capacity at least 1000 N and steel yield at least 200 MPa; both retain at least a 2x calculated static margin.

Power loss on a grade produces rollback in the negative control. Commission on a level surface with chocks; do not infer a fail-safe parking brake from a gearmotor.

Required physical validation

Bearing and pin fits; motor continuous torque/current/thermal behavior; gearbox axial thrust capacity; paired boom synchronisation; cable loop and direction-limit commissioning; print layer adhesion, deflection, creep and endurance. Record these before a general print or payload release.

Upper body, battery bay and independent motion



R3 upper body in the Cat 320 layout: cab envelope [120.0, 70.0, 118.0] mm at the machine's left front; measured tail radius 189.00 mm against 189 mm (2,830 mm at 1:15). Deck lowered 48 mm; exposed bridge/skirt gap 12 mm. Cab estimates use the Cat 320 page 4 drawing, not factory cab dimensions.

Battery bay check	Result
Cell envelope (Samsung 25R, LG HG2, Panasonic NCR18650B datasheets)	Ø18.0–18.6 × 64.7–65.3 mm, flat-top, unprotected
Welded 2S1P pack width / height / length	36.74–38.20 / 18.24–19.10 / 65.60–67.60 mm
Cradle grooves	Ø19.7 mm at 19.3 mm pitch, 69 mm between end walls
Minimum radial clearance	0.20 mm
End pad compressed (free 4 mm)	1.40–3.40 mm
Top pad compressed (free 3 mm)	1.33–2.09 mm
CAD intersection, largest / smallest pack	0 / 0 mm³
Pack lift-out 0–80 mm	65 exact intersections, 0 collisions
Pack to hatch-opening margin	14.4 mm
Upper-body screws checked	34; all engage the full nut; no tip below the Z124 deck underside

Flat-top unprotected 18650 cells only. Button-top or protected cells (about 67-70 mm) do not fit this cradle.

2S power budget: 36 W shared output, 6.2 V cutoff, 85% assumed efficiency: 6.83 A maximum. Required cells >=15 A, BMS >=10 A, pack controller limit 8 A. Thermal performance, current limits and runtime require hardware measurement.

CAD reference samples use boom stroke >=15 mm or slew within ±20°. This is not an operating restriction. The preview permits full 0–60 mm stroke and ±150° slew independently; operators manage track clearance.

Reference boundary tested at 666 configurations. The negative control outside this region has 6168.6 mm³ collision. The preview permits that pose; operators manage track clearance.